Machine Learning for Predicting API in Selangor

In 2023, Cheras recorded 31 days of unhealthy API readings

Problem Statements

CO
NO2
O3
PM2.5

- Respiratory diseases
- Cardiovascular conditions
- Premature mortality

Economic Hub

- Rapid urbanization
- Significant population density

Industrial Parks

Increased vehicular traffic

Objectives

- To identify the most effective ML models for predicting API levels in Selangor.
- To enhance environmental monitoring and raise public awareness about air quality issues.

Dataset

- Provided by JAS
- Air quality readings from 8 stations in Klang Valley
- Jan 1, 2020 to Dec 31, 2023

- 280495 rows
- 16 columns

Location | Date & Time | Air Quality Metrics | Meteorology

Data Preprocessing

Data Cleaning

- Check for duplicates
- Impute null values

Feature Engineering

- Extract hour, day, month, year
- Create lag intervals

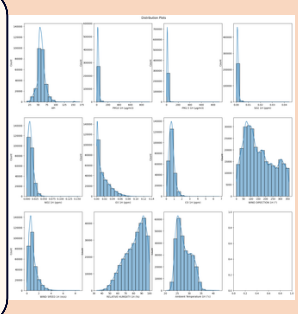
Feature Transformation

- Encode API category
- Scaling

Insights from Exploratory Data Analysis

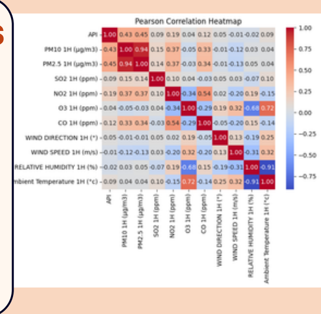
Univariate Analysis

- Most variables are positively skewed
- API outliers are kept to reflect real-world air quality patterns, like haze



Correlation Insights

- API moderately correlates with PM10 and PM2.5
- Pollutants show moderate correlation, suggesting shared sources like vehicles.



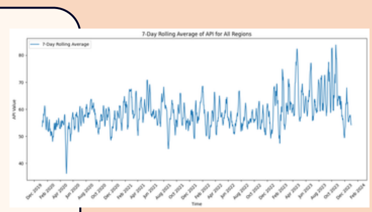
API: 7-Day Rolling Average

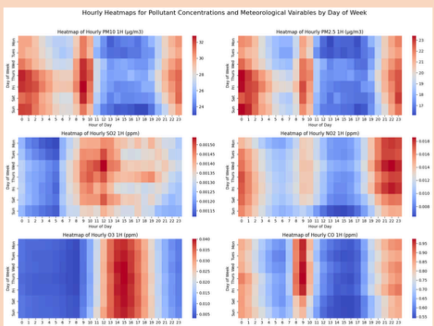
Sharp drop @ May 2020:

- Start of MCO
- Increased rainfall

Frequent spikes @ Sept 2023 :

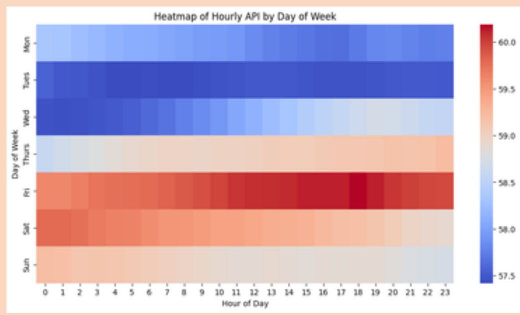
- Haze from peatland fires





Hourly Trends by Day

- Pollutant concentrations peak at 9 AM (morning rush hour) and 12 AM (low mixing layer and temperature).
- O_3 peaks at 3 PM due to increased photochemical formation under higher UV levels.
- API trends reflect the cumulative effect of pollutant concentrations, averaged over specific periods (e.g., 1-hour for NO_2, 8-hour for O_3, 24-hour for PM).



Models & Methods

Classification:

- Logistic Regression
- Naive Bayes
- Decision Tree
- Random Forest
- KNN

Regression

- Linear
- Lasso
- Decision Tree
- Random Forest

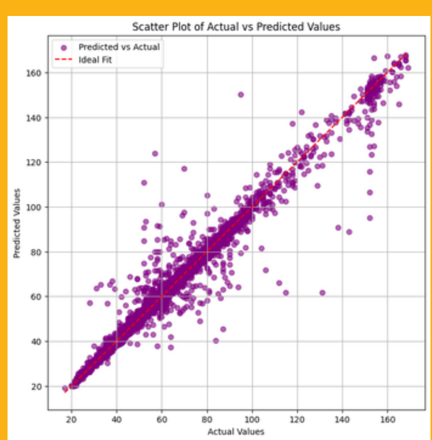
Time-series Forecasting

- LSTM

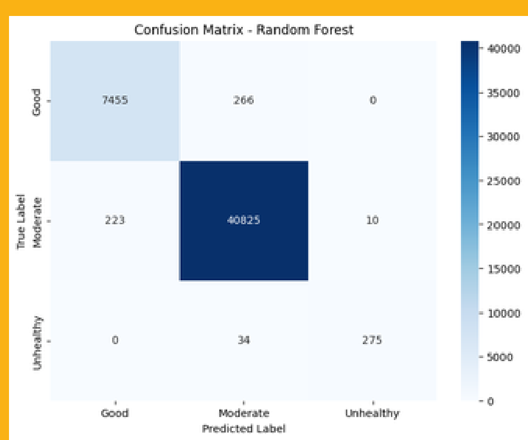
Results and Conclusion

- **Best Models:** XGBoost (regression) and Random Forest (classification) performed best.
- **Lag Features:** Temporal features, especially 1-hour API lag, improved accuracy.
- **LSTM Insights:** LSTM showed strong results but struggled with stability in recursive forecasting using longer windows, highlighted the trade-off between accuracy and stability.
- **Future Work:** Focus on optimizing model training for efficiency and exploring alternative forecasting methods.

XGBoost – Regression



Random Forest – Classification



LSTM – Recursive Forecast
CA16W Station (1h vs 24h)